

Julien Boussard

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Education

PhD, Columbia University Statistics Department

Sept 2019 - Sept 2024

- *Thesis*: Localizing spike sources for improved registration, spike-sorting, and decoding in large-scale Neuropixels recordings. Advised by Liam Paninski.
- *Research*: Developed data-driven approaches to analyze and decode large-scale electrophysiological recordings. Modelled real-world physical data using tools and techniques from signal processing, optimization, probabilistic modeling, machine learning, and dynamical systems.
- *PhD classes*: Machine Learning and Climate Change, Readings in Climate Science, Computational Statistics, Nonparametric Theory in Machine Learning, Statistical Inference Theory, Deep Probabilistic Models, Causal Inference, Causal ML, Neural Data Analysis, Computation and the Brain.

Statistics Master, Stanford University

Sept 2017 - March 2019

- *Studied Theoretical and Applied Statistics*: Theory of Probabilities, Modern Applied Statistics, Deep Learning, Reinforcement Learning, Introduction to Neuro-Engineering, Theoretical Models of the Neocortex.
- *Research Projects*: Decoding Neural Data with Recurrent Neural Network for Brain-Machine Interfaces, Risk-Sensitive Bayes-Adaptive Decision Making in Continuous Spaces, Predicting Subjective Sleep Quality from EEG signals.

Graduate Degree in Applied Mathematics, École Polytechnique

Sept 2014 - March 2017

- Studied Machine Learning, Statistics, Computer Science and Probabilities.

Classe Préparatoire MP*, Lycée Condorcet, Paris

Sept 2012 - July 2014

- Intensive two-year preparatory course in mathematics and physics for exams to French engineering schools.

Academic Experience

Postdoctoral researcher, McGill University, MILA

January 2025 - Present

- Under the supervision of Professor David Rolnick, I lead a project aiming at developing causal machine learning methods for improving robustness and physical consistency of climate emulators, and develop signal processing methods for segmenting and classifying bioacoustics recordings. and biodiversity monitoring.

Research Internship, MILA

June 2023 - September 2023

- Worked with Professor David Rolnick. Developed causal representation learning methods for discovering latent processes underlying climate model data.

Research Internship, Gatsby Computational Neuroscience Unit

June 2018 - September 2018

- Worked with Professor Maneesh Sahani. Developed a variational inference algorithm for learning latent stochastic differential equations from high-dimensional point-process observations to model neural dynamics.

Research Assistant, Stanford Intelligent Systems Laboratory

April 2018 - June 2018

- Worked with Professor Mykel Kochenderfer and Jamie Marc Zeitzer, Associate Professor at Stanford Center for Sleep Sciences and Medicine. Predicted Subjective Sleep Quality analyzing EEG data and derived sleep variables provided by the Sleep Heart Health Polysomnography dataset.

Publications

- [Identifiability Guarantees for Drivers and Dynamics of Delayed Physical Systems](#), J. Boussard, A. Debouchage, T. Saulus. *Submitted at ICLR, 2027*
- [2-Dimensional spectral gating for denoising bioacoustics recordings](#), J. Boussard, M. Teng, S. Saha, M. Gallego-Abenza. *Submitted at ICASSP, 2027*
- [Learning Hierarchical Causal Representations of the Effects of Forcings on Temperature in Climate Models](#), S. Zhao, I. Trajkovic, J. Kaltenborn, Y. Gurwicz, P. Nowack; D. Rolnick, J. Boussard. *Representations for the Physical Sciences Workshop, NeurIPS 2026 (Oral)*
- [Identifying birdsong syllables without labelled data](#), M. Teng*, J. Boussard*, D. Rolnick, H. Larochelle. *ICASSP, 2026*
- [Causal Climate Emulation with Bayesian Filtering](#), S. Hickman, I. Trajkovic, J. Kaltenborn, F. Pelletier, A. Archibald, Y. Gurwicz, P. Nowack, D. Rolnick, J. Boussard. *NeurIPS, 2025*
- [DREDge: robust motion correction for high-density extracellular recordings across species](#), C. Windolf, H. Yu, A. C. Paulk, D. Meszéna, W. Muñoz, J. Boussard, R. Hardstone, I. Caprara, M. Jamali, Y. Kfir, D. Xu, J. Chung, K. Sellers, Z. Ye, J. Shaker et al. *Nature Methods, 2024*
- [A modular approach to handle in-vivo drift correction for high-density extracellular recordings](#), S. Garcia, C. Windolf, J. Boussard, B. Dichter, A. P. Buccino, P. Yger, *eNeuro, 2024*
- [HybridBNN: Joint Training of Deterministic and Stochastic Layers in Bayesian Neural Nets](#), A. Nejatbakhsh, J. Boussard, *Workshop at the 6th Symposium on Advances in Approximate Bayesian Inference, 2024*
- [Towards Causal Representations of Climate Model Data](#), J. Boussard, C. Nagda, J. Kaltenborn, C. Lange, P. Brouillard, Y. Gurwicz, P. Nowack, D. Rolnick. *CCAI workshop, NeurIPS, 2023*
- [Ultra-high density electrodes improve detection, yield, and cell type specificity of brain recordings](#), Z. Ye, A. Shelton, J. Shaker, J. Boussard, J. Colonell, S. Manavi, S. Chen, C. Windolf, C. Hurwitz, T. Namima, F. Pedraja, S. Weiss, B. Raducanu, T. Ness, G. Einevoll et al. *Neuron, 2023*
- [DARTsort: A modular drift tracking spike sorter for high-density multi-electrode probes](#), J. Boussard, C. Windolf, C. Hurwitz, H. Lee, H. Yu, O. Winter, L. Paninski. *BiorXiv, 2023*
- [Bypassing spike sorting: Density-based decoding using spike localization from dense multielectrode probes](#), Y. Zhang, Tianxiao He, J. Boussard, C. Windolf, O. Winter, E. Trautmann, N. Roth, H. Barrell, M. Churchland, N. A. Steinmetz, E. Varol, C. Hurwitz, L. Paninski. *NeurIPS 2023 (spotlight)*
- [Robust Online Multiband Drift Estimation in Electrophysiology Data](#), C. Windolf, A. Paulk, Y. Kfir, E. Trautmann, D. Meszéna, W. Muñoz, I. Caprara, M. Jamali, J. Boussard, Z. Williams, S. Cash, L. Paninski, E. Varol. *ICASSP 2023*
- [Spike sorting pipeline for the International Brain Laboratory](#), International Brain Laboratory*, K. Banga, J. Boussard, G. A. Chapuis, M. Faulkner, K. D. Harris, J. M. Huntenburg, C. Hurwitz, H. D. Lee, L. Paninski, C. Rossant, N. Roth, N. A. Steinmetz, C. Windolf, O. Winter. *2022*
- [Three-dimensional spike localization and improved motion correction for Neuropixels recordings](#), J. Boussard*, E. Varol*, H. D. Lee, N. Dethe, L. Paninski. *NeurIPS 2021*
- [Decentralized Motion Inference and Registration of Neuropixel Data](#), E. Varol*, J. Boussard*, N. Dethe, O. Winter, A. Urai, The International Brain Laboratory, A. Churchland, N. Steinmetz, L. Paninski. *ICASSP 2021*
- [Learning interpretable continuous-time models of latent stochastic dynamical systems](#), L. Duncker, G. Böhner, J. Boussard, M. Sahani. *ICML 2019, Oral*
- [Predicting Subjective Sleep Quality Using Recurrent Neural Networks](#), J. Boussard, M. J. Kochenderfer, J. M. Zeitzer. *2019 IEEE Signal Processing in Medicine and Biology Symposium (Oral)*

Conference Presentations & Invited Talks

- Conference oral presentations
 - *Best practices and advice for using AI in research and writing*, Goldschmidt 2026 Conference
 - [Causal climate emulation](#), “Machine Learning for Climate Science” Session, EGU General Assembly 2025 (Highlight)
 - *Predicting Subjective Sleep Quality Using Recurrent Neural Networks*, 2019 IEEE Signal Processing in Medicine and Biology Symposium
- Conference & Workshop posters
 - [An AI-driven approach to enhancing wildfire representation and climate feedbacks in the UVic-ESCM v2. 10](#), “Spatial and Temporal Dynamics of Wildfires: Models, Theory, and Reality” Session, EGU General Assembly 2026
 - [Learning Causal Representations of Climate Model Data](#), “Advances in Emulating Earth System Models” Session, AGU General Assembly 2024
 - *Brain-wide Electrophysiological Atlas*, Cosyne 2024
 - [Towards Causal Representations of Climate Model Data](#), “Tackling Climate Change with Machine Learning” Workshop, NeurIPS 2023
 - *Density-based Neural Decoding using Spike Localization for Neuropixels Recordings*, Cosyne 2023
 - *Robust multiband drift estimation in electrophysiology data*, Cosyne 2023
 - *Decentralized motion inference and registration of Neuropixels data*, Cosyne 2021
- Invited Talks
 - *Causal Climate Emulation*, AI in Climate and Environmental Sciences Group Presentation, KIT, 2026
 - *Causal Climate Emulation*, ARCHES Group Presentation, INRIA Paris, 2025
 - *Learning Causal Representations of Climate Model Data*, LEAP Machine Learning Journal Club, Columbia University, 2024
 - *DARTsort: A modular drift tracking spike sorter for high-density multi-electrode probes*, Harris Lab, Janelia, 2023
 - *DARTsort: A modular drift tracking spike sorter for high-density multi-electrode probes*, Neurotheory Seminar, Center for Theoretical Neuroscience, Columbia University, 2023
 - *Three-dimensional spike localization and improved motion correction for Neuropixels recordings*, Neurotheory Seminar, Center for Theoretical Neuroscience, Columbia University, 2021

Teaching

Neural Data Analysis Lectures, Columbia University

Oct 2022, Oct 2023

- Gave two lectures on “Spike sorting: from basics to bleeding-edge” for Liam Paninski’s Neural Data Analysis PhD-level class, in 2022 and 2023.

Teaching Assistant, Columbia University, Statistics Department

September 2019 - June 2024

- I have been a teaching assistant for the following Statistics courses at Columbia:
 - UN1201: Calculus-based Intro to Statistics (Fall 2023, Spring 2024)
 - GU4241: Statistical Machine Learning (Spring 2023)
 - GR5205: Linear Regression Models (Fall 2022)
 - GU4001: Introduction to Probability and Statistics (Spring 2022)
 - W5701: Probability and Statistics for Data Science (Fall 2021)
 - UN1101: Introduction to Statistics (Spring 2021)
 - GR5242: Advanced Machine Learning (Fall 2020)
 - GU5206: Computational Statistics and Introduction to Data Science (Fall 2019, Spring 2020)

Teaching Assistant, Lycée Condorcet, Paris

Sept 2015 - March 2016

- Gave mathematics lessons for students preparing the exam for French engineering schools.

Funding

- Bourse Postdoctorale Fonds de recherche du Québec Nature et technologies 2026-2028, 90,000 CAD
- Programme Projets Exploratoires IVADO 2025-2026, 100,000 CAD

Service

- Area Chair for ICLR 2027
- Reviewers for journals, conferences, and workshops
 - Reviewer for ICLR, ICML, NeurIPS, ICASSP (2021–2026)
 - Top / Gold reviewer at NeurIPS 2024, NeurIPS 2025, ICML 2026
 - Reviewer for NeurIPS CCAI workshop (2023–2025), NeurIPS RPS workshop (2026)
 - Reviewer for IEEE Transactions on Signal Processing, Transactions in Machine Learning Research (TMLR), Nature Communications, Machine Learning: Earth, Science Advances
- Advisor for the Frontier Development Lab Research Sprint 2025, *Global 3D Reconstruction of Clouds & Tropical Cyclones*.

Engagement & Public Outreach

- Co-organizer of the [Critical Science Reading Group at Mila](#), 2025-present
- Interview for Radio-Canada (National TV), How is AI used for Weather Predictions?, April 2026

Non-Academic Experience

Software Engineering Internship, STOIC, New-York

March 2017 - August 2017

- Part of the Research and Development team. Implemented various distributed Machine Learning algorithms in Javascript and C++, such as logistic regression, k-means clustering and support vector machines. The STOIC users can use these algorithms to run fast machine learning analysis on their own large datasets.

Summer Internship, Société Générale, New-York

June 2016 - August 2016

- Summer intern analyst in the Flow Strategies and Solutions team. Developed weekly quantitative and macroeconomic trading strategies using time-series and statistical analysis. Created several VBA pricing functions for derivative products. Developed a method to determine the best trades on ETFs.

Order of Malta, Paris

Sept 2014 - June 2016

- Work experience at the Headquarters of First Aid and Social Assistance of the Order of Malta. Organized first aid events and helped the Paris Fire Brigade as a first aider. Volunteered in an emergency shelter for homeless people. Gave French lessons to undocumented migrants.